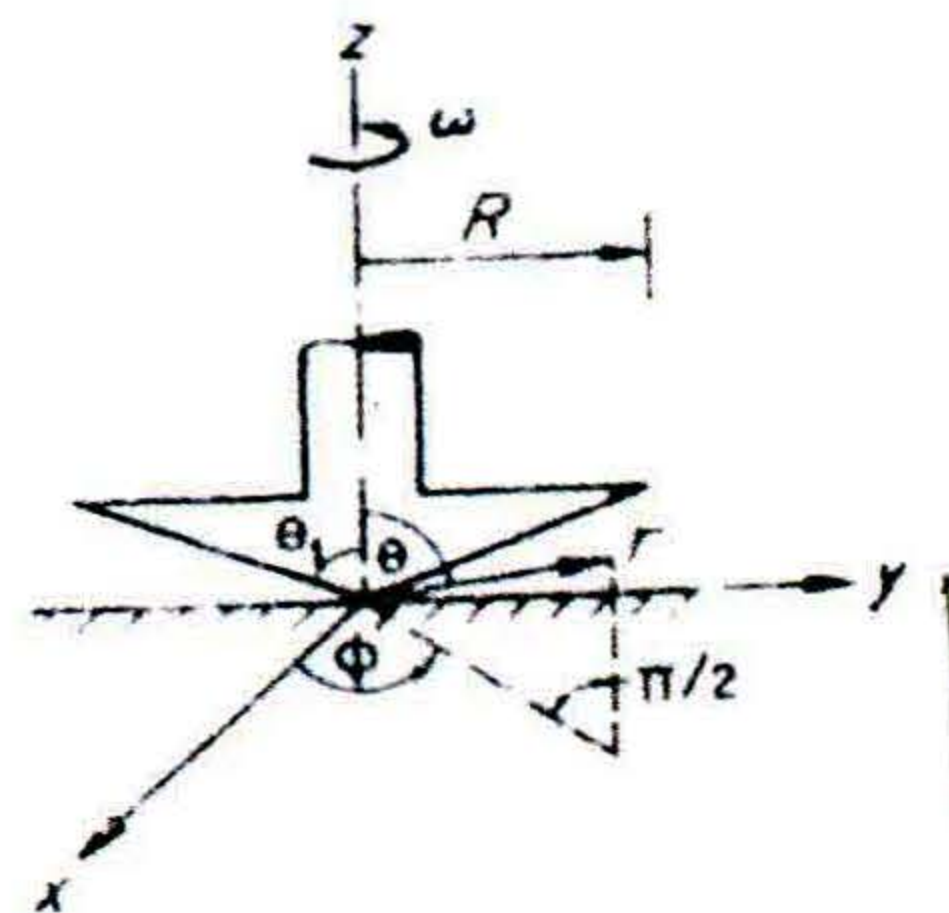


A cone and plate viscometer consists of a cone rotating at a constant angular velocity ω over a stationary plate. Liquid fills the gap between the cone and plate and the torque required to keep the plate stationary is measured through the deflection of a tensional spring. Simplify the Navier-Stokes equation assuming that flow is laminar and tangential (i.e., only $v_\phi \neq 0$), and that the inertial and gravity terms are negligible. Assuming $v_\phi = rf(\theta)$, show that $f(\theta)$ must satisfy

$$f'' + f' \cot \theta + f \left(2 - \frac{1}{\sin^2 \theta} \right) = 0$$

If the cone angle $[\pi/2 - \theta_1]$ is small, the velocity V_ϕ may be fairly well approximated as linear in the vertical direction. Compute the torque on the lower plate in terms of μ , R , ω , and θ_1 . If $R=5\text{cm}$, $\theta_1=89^\circ$, torque $= 5 \times 10^{-3} \text{ Nm}$, $\omega = 2 \text{ rpm}$, determine the viscosity of the liquid.



For this case $v_r = v_\theta = 0$, $v_\phi \neq 0$.

By symmetry $\frac{\partial}{\partial \phi} = 0$.

Inertial and gravity terms of N.S. $\rho \mathbf{a}^n$ are negligible.

$\therefore r$ comp. $\Rightarrow \frac{\partial p}{\partial r} = 0$. θ comp. $\Rightarrow \frac{\partial p}{\partial \theta} = 0$.

$$\phi: \Rightarrow \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_\phi}{\partial \theta} \right) - \frac{v_\phi}{\sin^2 \theta} = 0.$$

$v_\phi = r f(\theta)$; substitute in the above $\rho \mathbf{a}^n$.

$$\frac{\partial}{\partial r} (r^2 f) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cdot r f') - \frac{r f}{\sin^2 \theta} = 0.$$

$$2rf + \frac{r}{\sin \theta} [\cos \theta f' + f' \sin \theta] - \frac{r f}{\sin^2 \theta} = 0.$$

$$2f + f' \cot \theta + f'' - \frac{f}{\sin^2 \theta} = 0.$$

$$\therefore f'' + f' \cot \theta + f \left(2 - \frac{1}{\sin^2 \theta} \right) = 0$$

If the cone angle is small

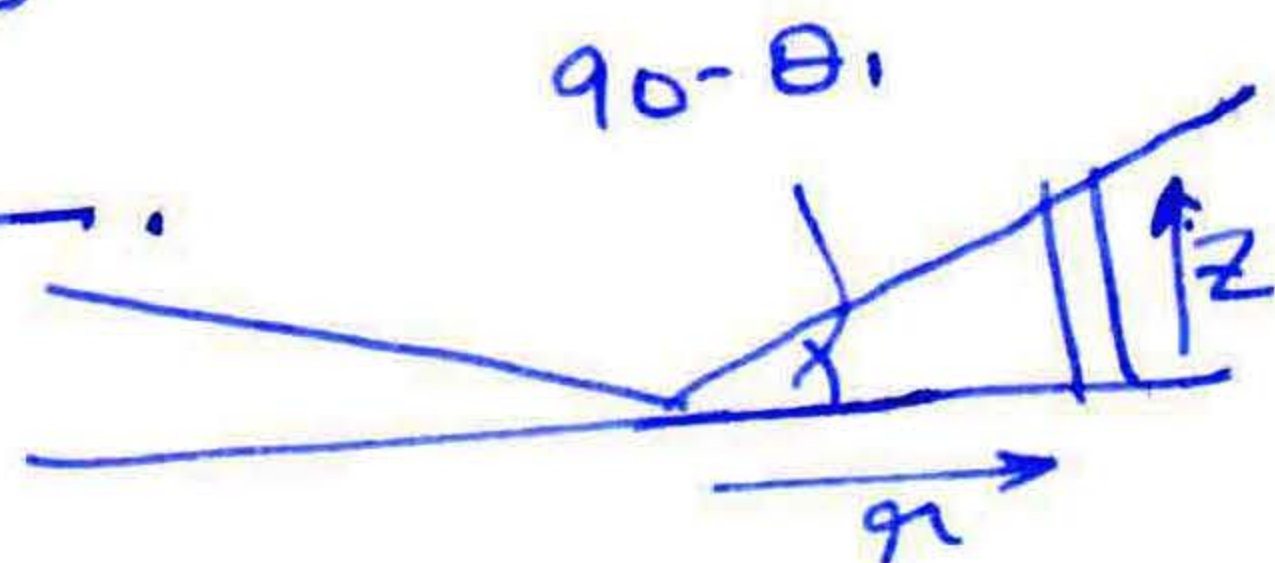
$$v_\phi = c_1 z + c_2 \quad (\text{linear})$$

$$z=0, v_\phi=0 \Rightarrow c_2=0, \text{ and } z = r \cot \theta_1,$$

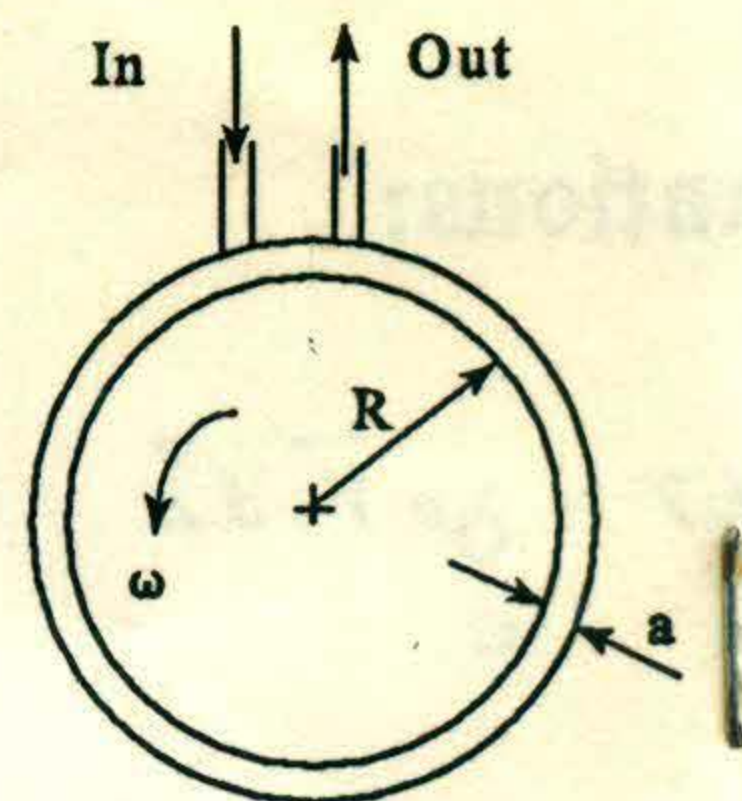
$$v_\phi = \omega r = \frac{\omega z}{\cot \theta_1} \Rightarrow \tau_{z\phi} = \frac{\mu \omega}{\cot \theta_1}.$$

$$\therefore \text{Torque on lower plate} = \int_0^R \tau_{z\phi} \times 2\pi r dr \times r$$

$$T = \frac{\mu \omega}{\cot \theta_1} \frac{2\pi R^3}{3} \therefore \mu = \frac{T \cot \theta_1 \times 3}{\omega \cdot 2\pi R^3} = 1.6 \text{ Pa-s}.$$



A viscous shear pump is made from a stationary housing with a close-fitting rotating drum inside. The clearance is small compared to the diameter of the drum. Fluid is dragged around the annulus by viscous forces. Evaluate the performance characteristics of the shear pump (pressure differential, input power and efficiency) as functions of volumetric flow rate. Assume that the depth normal to the diagram is b .



$$0 = -\frac{dP}{dx} + \mu \frac{d^2 v_x}{dy^2}$$

$$v_x = \frac{1}{2\mu} \left(\frac{dP}{dx} \right) y^2 + c_1 y + c_2 \quad (1)$$

BC $y=0 \quad v_x=0 \Rightarrow c_2=0.$

$y=a \quad v_x=\omega R \Rightarrow c_1 = \frac{1}{a} \left[\omega R - \frac{1}{2\mu} \left(\frac{dP}{dx} \right) a^2 \right]$

$$v_x = \frac{1}{2\mu} \frac{dP}{dx} a^2 \left[\left(\frac{y}{a} \right)^2 - \left(\frac{y}{a} \right) \right] + \frac{\omega R}{a} y \quad (2)$$

$$\langle v_x \rangle = \frac{1}{ab} \int_0^a \int_0^b v_x dy dz = \frac{1}{a} \int_0^a v_x dy = -\frac{1}{12\mu} \frac{\Delta P}{L} a^2 + \frac{\omega R}{2}$$

$$Q = ab \langle v_x \rangle, \quad \frac{Q}{b} = -\frac{1}{12\mu} \frac{\Delta P}{L} a^3 + \frac{\omega R a}{2}$$

$$\therefore \Delta P = \frac{6\mu L \omega R}{a^2} \left[1 - \frac{2Q}{ab\omega R} \right] \quad (3)$$

$$\tau|_{y=a} = +\mu \left. \frac{dv_x}{dy} \right|_{y=a} = \mu \left[\frac{1}{\mu} \frac{\Delta P}{L} y - \frac{1}{2\mu} \frac{\Delta P}{L} a + \frac{\omega R}{a} \right] \Big|_{y=a}$$

$$= \frac{\Delta P}{L} a - \frac{\Delta P}{L} \frac{a}{2} + \frac{\omega R}{a} \mu = \frac{\Delta P}{L} \frac{a}{2} + \frac{\omega R \mu}{a}$$

$$= \frac{3\mu L \omega R}{aL} \left[1 - \frac{2Q}{ab\omega R} \right] + \frac{\omega R \mu}{a}$$

Torque = $\tau(Lb) \cdot R$, Power (P) = Torque $\times \omega = \tau \cdot \omega R L b$

$$P = \frac{\mu L b (\omega R)^2}{a} \left[4 - \frac{6Q}{ab\omega R} \right]$$

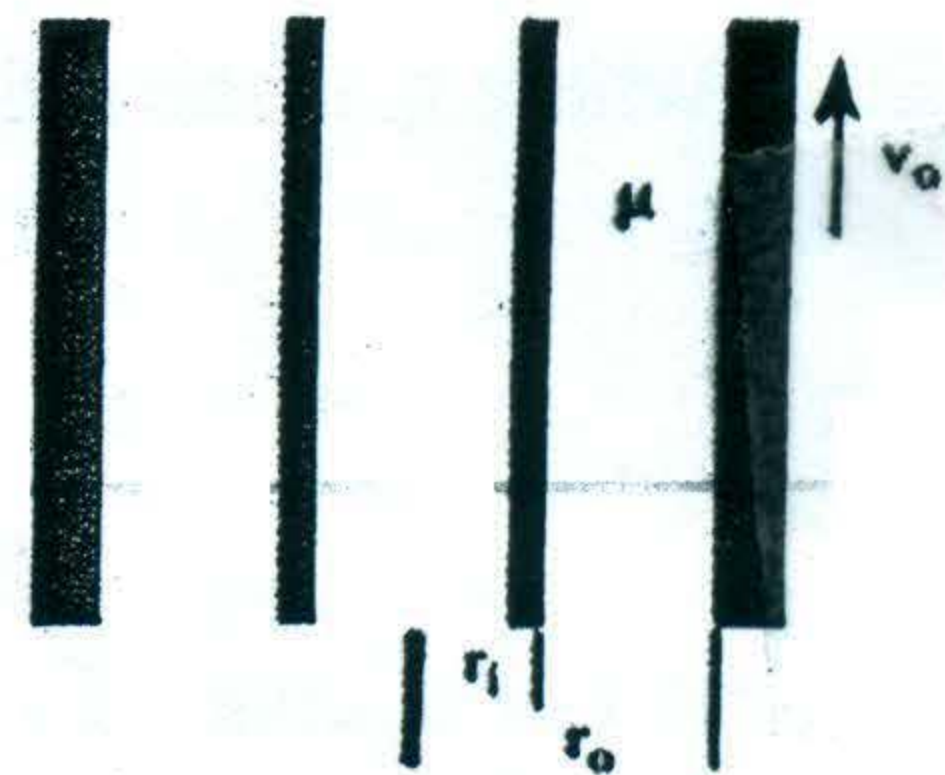
Output power = $Q \Delta P$ $\therefore$ Efficiency, $\eta = \frac{Q \Delta P}{P(\text{power})}$

$$= \frac{6Q}{ab\omega R} \frac{\left(1 - \frac{2Q}{ab\omega R} \right)}{\left(4 - \frac{6Q}{ab\omega R} \right)}$$

The space between two very long concentric cylinders is filled with water. The inner cylinder is stationary while the outer cylinder is pulled along axially with a velocity V_0 .

a) Starting with the relevant governing equation and boundary conditions evaluate the velocity profile and the stress at the surface of the inner cylinder.

b) What is the volumetric flow rate of water at any axial location?



Starting and simplifying NS eqⁿ in cylindrical coordinates (cancelling the terms which are not relevant)
 i.e. no stress in θ, z , $\frac{dv_{\theta z}}{d\theta} = 0$, $\frac{dv_z}{dz} = 0$
 no vel. in θ, r ($v_{\theta} = v_r = 0$)

S.S.

$$-\frac{d}{dr}(A \tau_{rz}) = 0.$$

$$\tau_{rz} = -\mu \frac{dv_z}{dr}, \quad A = 2\pi r L$$

$$\therefore 2\pi\mu L \frac{d}{dr}(r \frac{dv_z}{dr}) = 0 \quad \text{--- (1)}$$

B.C. $r = r_i, \quad v_z = 0.$
 $r = r_o, \quad v_z = V_0.$

Integrating (1) $v_z = C_1 \ln r + C_2$

Applying the B.C.s.

$$v_z = \frac{V_0}{\ln(r_o/r_i)} \ln(r/r_o) + V_0.$$

Shear stress. $\tau_{rz} = -\mu \frac{dv_z}{dr} = -\frac{\mu V_0}{\ln(r_o/r_i)} \frac{1}{r}.$

The volumetric flow rate

$$Q = 2\pi L \int_{r_i}^{r_o} v_z dr = \frac{V_0}{\ln(r_o/r_i)} \int_{r_i}^{r_o} \ln \frac{r}{r_o} \cdot dr + V_0(r_o - r_i)$$

$$\begin{aligned} v_0 &= C_1 \ln r_o + C_2 \\ 0 &= C_1 \ln r_i + C_2 \\ \hline C_1 \ln \frac{r_o}{r_i} &= V_0. \\ C_1 &= \frac{V_0}{\ln(r_o/r_i)} \\ C_2 &= V_0 - C_1 \ln r_o \\ &= V_0 - \frac{V_0}{\ln(r_o/r_i)} \ln r_o \end{aligned}$$